Chapter 7, Problem 84P

Problem

A capacitor with a value of 10 mF has a leakage resistance of 2 MΩ. How long does it take the voltage across the capacitor to decay to 40% of the initial voltage to which the capacitor is charged? Assume that the capacitor is charged and then set aside by itself.

Step-by-step solution

Step 1 of 4

Draw a circuit having capacitor with its leakage resistance.

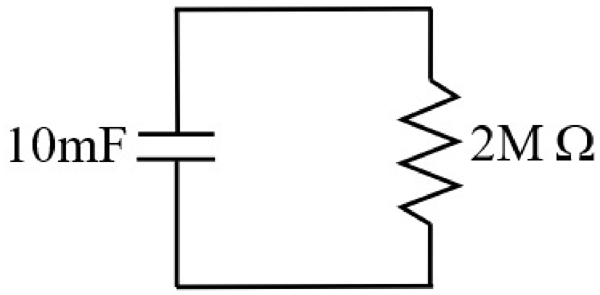

Figure 1

Step 2 of 4

The voltage across the capacitor is,

$$v_C(t) = v_C(\infty) + \left(v_C(0) - v_C(\infty)\right)e^{\frac{-t}{\tau}} \quad \ldots\ldots (1)$$

The final voltage across the capacitor $v_C(\infty)$ is,

$$v_C(\infty) = 0\ \text{V}$$

Then, rewrite the equation (1).

$$v_C(t) = \left(v_C(0)\right)e^{\frac{-t}{\tau}} \quad \ldots\ldots (2)$$

Step 3 of 4

The voltage across the capacitor decay to 40%. Then,

$$\begin{aligned} v_C(t) &= \frac{40}{100}v_C(0) \\ &= 0.4v_C(0) \end{aligned}$$

Calculate the time constant τ.

$$\begin{aligned} \tau &= RC \\ &= 20 \times 10^3 \end{aligned}$$

Step 4 of 4

Substitute $0.4v_C(0)$ for $v_C(t)$ and 20×10^3 for τ in equation (2).

$$0.4v_C(0) = v_C(0)e^{-0.5\times10^{-3}t}$$
$$0.4 = e^{-0.5\times10^{-3}t}$$

Take the natural log.

$$0.4v_C(0) = v_C(0)e^{-0.5\times10^{-3}t}$$
$$0.4 = e^{-0.5\times10^{-3}t}$$
$$t = 1832\text{ sec}$$

Hence, the capacitor takes $\boxed{1832\text{ sec}}$ to decay to 40% of the initial value.